

Md Kamrul Islam

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)



RESEARCH INTEREST

Natural Language Processing, Large Language Models, Information Extraction, Retrieval-Augmented Generation, Deep Learning, and Data Mining.

EDUCATION

Erasmus Mundus Joint Master's Degree in Big Data Management and Analytics (BDMA)

CentraleSupélec, Université Paris-Saclay | UPC, Barcelona | ULB, Brussels

Weighted Average: 14.5/20

Mobility Path:

- **Université libre de Bruxelles (ULB)**, Brussels, Belgium
- **Universitat Politècnica de Catalunya (UPC)**, Barcelona, Spain
- **CentraleSupélec, Université Paris-Saclay**, Paris, France

Master's Thesis: SecBPMN2 Assistant – A Hybrid LLM-Based Framework for Automating Security Annotations in Business Process Models.

Bachelor of Engineering in Software Engineering

Sichuan University

CGPA: 3.64/4.00 (Weighted Average: 87%)

Thesis: Brain Tumor Detection and Classification using Convolutional Neural Networks.

Sept 2023 – Oct 2025

France | Spain | Belgium

Mar 2018 – Jan 2022

Chengdu, China

WORK EXPERIENCE

Research Engineer Intern

LISSI, Université Paris-Est Créteil (in collaboration with CEA-LIST, France)

May 2025 – Oct 2025

Créteil, France

- Proposed a novel hybrid framework integrating LLM-based reasoning and rule-based validation for automated cybersecurity-annotated BPMN model generation.
- Optimized annotation accuracy and efficiency through advanced prompt engineering and retrieval-augmented generation (RAG) across multiple LLMs.
- Constructed a benchmark dataset and evaluation framework supporting quantitative and qualitative metrics.
- Achieved higher annotation accuracy while reducing annotation time by 95% compared to domain experts.

Tech Stack: Python, LangChain, LLMs, RAG, Prompt Engineering, React.js, Flask, CI/CD

Graduate Research Assistant | [Project Link](#)

Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), CentraleSupélec

Oct 2024 – Feb 2025

Gif-sur-Yvette, France

- Developed an unsupervised deep-learning framework with Group-Equivariant CNNs to NIH Chest X-ray dataset.
- Improved clustering and feature extraction by encoding geometric symmetries, reducing reliance on data augmentation.
- Implemented distributed training pipelines for scalable, high-performance experiments with early disease detection.

Tech Stack: PyTorch, OpenCV, HPC, CUDA, LaTeX

Software Engineer Intern

Chengdu Suncap Co., Ltd

Dec 2020 – May 2021

Chengdu, China

- Developed end-to-end data pipeline for large-scale data ingestion, transformation, and analytics in distributed environments.
- Enhanced data preprocessing and feature-engineering, boosting predictive model accuracy by approximately 20%.
- Collaborated within an Agile Scrum environment to implement best practices in modular design, version control, and continuous integration.

Tech Stack: Apache Spark, Scikit-learn, SQL, Agile Scrum, Jira, Git, CI/CD

TECHNICAL SKILLS

Programming & Framework: Python, C/C++ (OOP), SQL, JavaScript, React.js, Flask
Data Science & ML: PyTorch, Scikit-learn, LangChain, Hugging Face, OpenCV, CUDA
Big Data & Cloud: PySpark, Hadoop, Apache Airflow, Microsoft Fabric, Microsoft Azure, HPC
Graph & Database Systems: PostgreSQL, Neo4j, GraphDB, Cypher, SPARQL, ETL/ELT Pipelines
Data Visualization: Power BI, Tableau, Streamlit, D3.js
Tools & Collaboration: Linux, Git, Overleaf Latex, Docker, Agile Scrum, CI/CD

PROJECTS

DigiScan360 Project Link <i>Developed a competitive intelligence platform and pitched it as a startup prototype at UPC's entrepreneurship initiative.</i> Tech Stack: PySpark, LLMs, Microsoft Fabric, Azure Data Factory, Power BI, GraphDB, SPARQL	Feb 2024 – June 2024
Anomaly Detection in Diesel Train Cooling Systems Project Link <i>Developed unsupervised models to detect anomalies in train cooling systems for the Belgian National Railway Company (SNCB).</i> Tech Stack: Python, SQL, Scikit-learn, Tableau	Sept 2023 – Dec 2023
PostgreSQL Extension for Chess Game Analysis Project Link <i>Created a custom PostgreSQL extension for storing and analyzing chess games.</i> Tech Stack: PostgreSQL, C, SQL, Linux, Database Systems, Indexing	Sept 2023 – Dec 2023
Brain Tumor Detection and Classification by Using CNN Project Link <i>Achieved 98% accuracy in brain tumor detection using a custom CNN model.</i> Tech Stack: TensorFlow, Keras, Python, OpenCV	Sept 2021 – Dec 2021
AI-Based Disease Prediction System Project Link <i>Built a web-based symptom checker predicting over 40 diseases with machine learning algorithms.</i> Tech Stack: Python, Scikit-learn, Django, ReactJS, REST APIs	Mar 2021 – June 2021

PUBLICATIONS

C. K. Sah, L. Xiaoli, M. M. Islam and M. K. Islam , “Navigating the AI Frontier: A Critical Literature Review on Integrating Artificial Intelligence into Software Engineering Education,” 2024 36th International Conference on Software Engineering Education and Training (CSEE&T), Würzburg, Germany, 2024, pp. 1–5. doi:10.1109/CSEET62301.2024.10663054

EXTRA-CURRICULAR ACTIVITIES

Twelfth European Big Data Management & Analytics Summer School (eBISS 2024) University of Padova Poster Link Paper Link	Padova, Italy July 2024
• Participated in the fully funded summer school sponsored by the European Commission. • Presented a research poster titled “Applying Knowledge Graphs in Retrieval-Augmented Generation (RAG)”.	

SCHOLARSHIPS

• Received the Erasmus Mundus Partner Country Scholarship (2023–2025), fully funded by the European Commission. • Awarded the Belt and Road Initiative Scholarship (2018–2022), full funding by the Chinese Government.	
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LANGUAGES

English: C2 (Proficient) Chinese: B2 (Upper-intermediate) French: A2 (Elementary)
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REFERENCES

Tiphaine HENRY, PhD Researcher, LICIA Laboratory, CEA LIST Paris, France Email: tiphaine.henry@cea.fr Master's Thesis Supervisor	Mahmoud SAKR Professor, Université Libre de Bruxelles Brussels, Belgium Email: mahmoud.sakr@ulb.be Graduate Course Instructor
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